

ML PROJECT

We have chosen the topic to predict the food delivery time after ordering the food. The columns in the dataset numbers but in the form of string. Initially, we 20 column and those are :-

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[ 'ID', 'Delivery_person_ID', 'Delivery_person_Age', 'Delivery_person_Ratings',  
'Restaurant_latitude', 'Restaurant_longitude', 'Delivery_location_latitude',  
'Delivery_location_longitude', 'Order_Date', 'Time_Orderd', 'Time_Order_picked',  
'Weatherconditions', 'Road_traffic_density', 'Vehicle_condition', 'Type_of_order',  
'Type_of_vehicle', 'multiple_deliveries', 'Festival', 'City', 'Time_taken(min)' ]
```

Time_taken(min) column was like this:-

Time_taken(min)
(min) 24
(min) 33
(min) 26
(min) 21
(min) 30
...
(min) 32
(min) 36
(min) 16
(min) 26
(min) 36

Fig 1: to show the feature 'Time_taken(min)'

As you could see in the above picture that feature was in the form of string so after applying one-hot-encoding, several features like 'Time_taken(min)_(min)24', 'Time_taken(min)_(min)33', 'Time_taken(min)_(min)26' etc were made. So, We have to convert the 'Time_taken(min)' feature into numbers. Now, the 'Time_taken(min)' feature only contains min in the form of numbers.

We choose to apply random forest first because it deals with means. Then after applying random forest on the data, we get the result:-

	MSE	MAE	R ²
Random Forest	24.31687176225463	3.858326570895932	0.7226575213673818



Fig 2: graph to show the Actual Delivery Time vs Predicted Delivery Time by Random Forest

Then, we applied ANN with the dropout of 0.2 in all the layers and the result was:-

				MSE	MAE	R ²
128	64	32	1	30.875873565673828	4.395723252537507	0.6478561925201282
64	32	16	1	32.38364791870117	4.488470516859511	0.630651229170132

32	16	8	1	40.801498 41308594	5.0170044 02521239 5	0.5346358 14736780 9
256	128	16	1	30.927600 86059570 3	4.4031662 38094764	0.6472619 74010671 5

Despite of trying ANN multiple times, we are unable to reach the R^2 above 0.64 so, we switched to linear regression and the output was:-

	MSE	MAE	R^2
Linear Regression	30.11583728882478 2	4.352744343766918	0.656518279101009 3

Linear Regression also gives us the R^2 of 0.65.

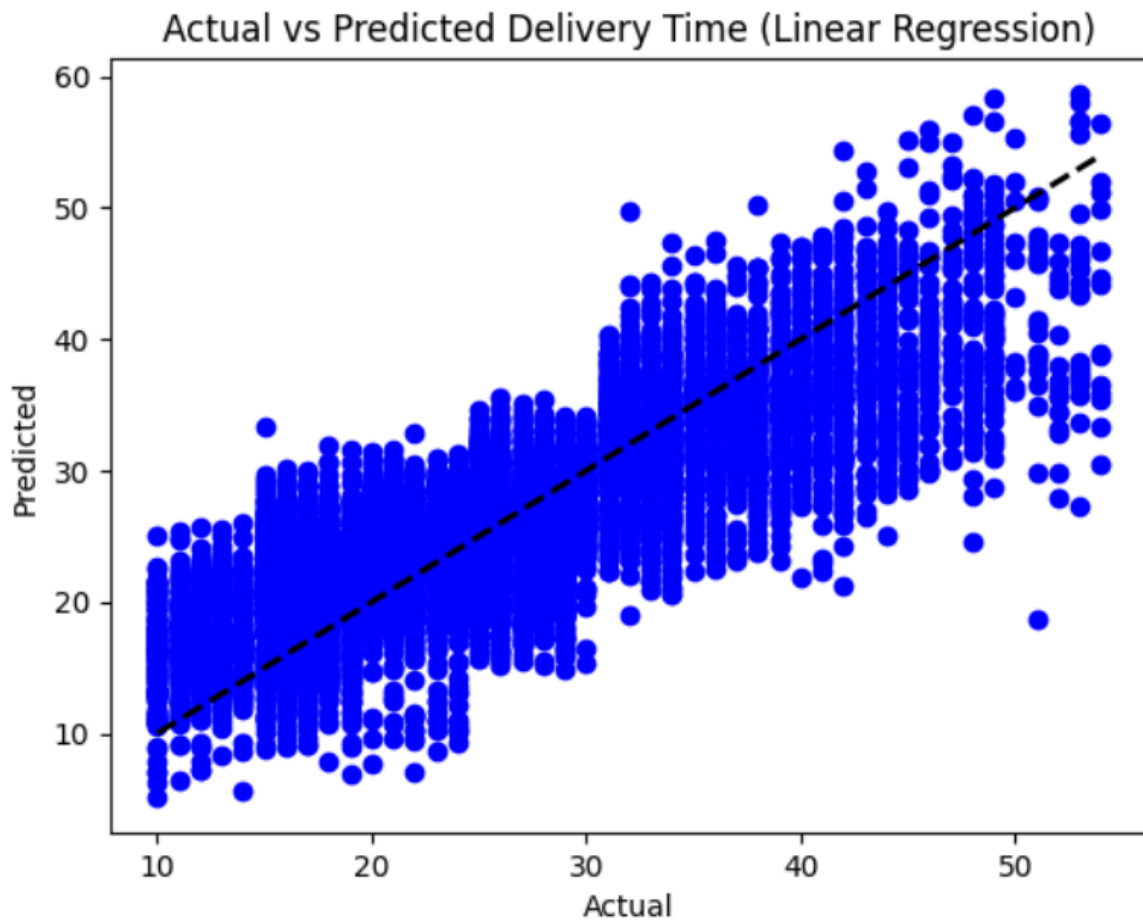


Fig 3: graph to show the Actual Delivery Time vs Predicted Delivery Time by Linear Regression

After trying all the algorithms, we got the maximum R^2 by Random forest which is 0.72 so, we decided to clean the data.

We found that the number of features suddenly increased to 47446. But we only had 20 features in the beginning.

[10]:

	Restaurant_latitude	Restaurant_longitude	Delivery_location_latitude	Delivery_location_longitude
0	22.745049	75.892471	22.765049	75.912471
1	12.913041	77.683237	13.043041	77.813237
2	12.914264	77.678400	12.924264	77.688400
3	11.003669	76.976494	11.053669	77.026494
4	12.972793	80.249982	13.012793	80.289982
...	...	...	...	...
45588	26.902328	75.794257	26.912328	75.804257
45589	0.000000	0.000000	0.070000	0.070000
45590	13.022394	80.242439	13.052394	80.272439
45591	11.001753	76.986241	11.041753	77.026241
45592	23.351058	85.325731	23.431058	85.405731

45593 rows × 47446 columns

Fig 4: to show the number of columns

After trying multiple times we found that the 'ID' column was unique. So, because of one hot encoding, a feature of every particular id was made which increases the number of columns. So, we dropped the number 'ID' column. The number of features left were 126. Now, Take a look at the time column which were 'Time_Orderd' and 'Time_Orderd_picked'.

Time_Orderd	Time_Order_picked
11:30:00	11:45:00
19:45:00	19:50:00
08:30:00	08:45:00
18:00:00	18:10:00
13:30:00	13:45:00
...	...
11:35:00	11:45:00
19:55:00	20:10:00
23:50:00	00:05:00
13:35:00	13:40:00
17:10:00	17:15:00

Fig 5: to show the previous time columns

It is in the form HH : MM : SS. So, we split 'Time_Orderd' in the form 'Time_Orderd_Hour', 'Time_Orderd_Minute' and 'Time_Orderd_Second'.

Time_Orderd_Hour	Time_Orderd_Minute	Time_Orderd_Second
11.0	30.0	0.0
19.0	45.0	0.0
8.0	30.0	0.0
18.0	0.0	0.0
13.0	30.0	0.0
...	...	...
11.0	35.0	0.0
19.0	55.0	0.0
23.0	50.0	0.0
13.0	35.0	0.0
17.0	10.0	0.0

Fig 5: to show split with the newest time columns

Same thing with the feature 'Time_Order_picked'. So, after applying EDA on Time feature, the result by applying Random Forest was :-

	MSE	MAE	R ²	Number of features left
Random forest after EDA on time column	23.1927407391 1613	3.78434696786 92845	0.73547863122 53114	130

Though we won't get the satisfied result, the features 'Delivery_person_Age' were in the form of string so we converted them in the form of Integer. Also, in the NaN values, we filled the mean of all the values of the same column.

	MSE	MAE	R ²	Number of features left
Random forest after EDA on age column	20.599395789011954	3.5700142559491175	0.7650566428808941	109

Then, we changed the feature 'Delivery_person_Ratings' from String to Integer. Also, in the NaN values we filled the mean of all values of the same column. Then, the result was:-

	MSE	MAE	R ²	Number of features left
Random forest after EDA on rating	20.36376079613993	3.536507292466279	0.7677441428856089	82

Then, we again changed the feature 'multiple_deliveries' from String to Integer. Also, in the NaN values we filled the mean of all values of the same column. Then, the result was:-

	MSE	MAE	R ²	Number of features left
Random forest after EDA on multiple deliveries	20.42954501590087	3.5452330299374935	0.7669938507122656	79

After being unsuccessful in this attempt also, then we tried Linear Regression and the result was:-

	MSE	MAE	R ²	Number of features left
LR after EDA	38.173893142697985	4.879424789091879	0.5646133167636145	79

Then we even tried ANN with the dropout of 0.2 in all the layers and the result was:-

				MSE	MAE	R ²
256	128	16	1	32.820144 65332031	4.4986233 99831757	0.6256781 92177510 1

END

	mse	mae	R2		
Decision tree	44.89209343 129729	5.043425814 234017	0.487989878 592481		

	mse	mae	r2	
Random forest after EDA on time column	23.1927407391 1613	3.78434696786 92845	0.73547863122 53114	130
Random forest after EDA on age column	20.5993957890 11954	3.57001425594 91175	0.76505664288 08941	109
Random forest after EDA on rating	20.3637607961 3993	3.53650729246 6279	0.76774414288 56089	82
Random forest after EDA on multiple deliveries	20.4295450159 0087	3.54523302993 74935	0.76699385071 22656	79
LR after EDA	38.1738931426 97985	4.87942478909 1879	0.56461331676 36145	
Ann after EDA	32.8201446533 2031	4.49862339983 1757	0.62567819217 75101	